Portable ROS 2 Sensor-Fusion Architecture for Autonomous Rover Indoor 3D Mapping

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***Abstract*—A sensor fusion architecture based on graph-based SLAM (simultaneous localization and mapping) for 3D reconstruction of indoor environments is presented. The system integrates data from a 2D LiDAR sensor and an RGB-D camera within the ROS 2 ecosystem, using RTAB-Map to perform pose-graph optimization and loop-closure detection. The proposed architecture operates seamlessly in both simulation and physical robotic platforms without requiring modifications to its core implementation. The quantitative geometric evaluation was conducted in a digital-twin environment of a military transport aircraft. The Cloud-to-Mesh (C2M) distance metrics, obtained from a single simulated run, yielded a mean reconstruction error of 0.0308 m and a standard deviation of 0.0439 m, corresponding to approximately 0.115% of the dimensions of the reference structure. Residual analysis indicated an approximately Gaussian error distribution without predominant systematic components, demonstrating the effectiveness of the global optimization in reducing odometric drift within the simulated environment. Deployment on the physical robotic platform confirmed functional and operational portability and continuous execution, although quantitative real-world reconstruction accuracy was not evaluated in this work.**

***Index Terms*—3D reconstruction, graph-based SLAM, RTAB-MAP, ROS 2, sensor fusion**

I. Introduction

Three-dimensional reconstruction has become a key component of modern robotic perception [1]. In geometrically constrained indoor environments [2], [3], generating consistent real-time spatial representations is essential not only for autonomous navigation and path planning [4], [5] but also for creating high-fidelity digital twins supporting inspection, maintenance, and virtual commissioning [6].

Recent SLAM advances have improved localization accuracy and environmental reconstruction. However, single-sensor configurations frequently degrade in performance under illumination changes, textureless scenes, limited sensing range, or insufficient geometric information, highlighting the need for multimodal fusion architectures that exploit complementary sensing characteristics [4].

Although several SLAM frameworks integrating RGB-D cameras and LiDAR have been proposed, most focus on algorithmic improvements, sensor fusion strategies, or platform-specific evaluations [5]. Few studies investigate the portability of a complete ROS 2-based SLAM architecture capable of operating without software modifications in both simulation and embedded robotic systems, while simultaneously providing rigorous quantitative validation of the reconstructed geometry.

To address this gap, this paper proposes a portable, ROS 2-native sensor fusion architecture based on RTAB-Map that integrates RGB-D and 2D LiDAR data for real-time indoor 3D reconstruction. RTAB-Map's native support for heterogeneous sensor fusion, pose graph optimization, and memory management makes it well-suited for resource-constrained platforms [7]. Unlike existing implementations, the proposed framework preserves an identical software architecture in both simulation and physical deployment, enabling direct migration and systematic validation through a digital twin reference model. The architecture combines 2D LiDAR scan matching, 3 DoF pose graph optimization, and RGB-D data for dense 3D reconstruction, providing a portable framework for consistent indoor mapping.

To validate this architecture, this work adopts a digital twin methodology that enables seamless migration from high-fidelity simulation to a physical mobile robot [8]. A quantitative evaluation framework assesses reconstruction accuracy by geometrically comparing reconstructed point clouds with the ground truth digital twin reference model through Cloud-to-Mesh (C2M) signed distance analysis.

The main contributions are summarized as follows:

- **Portable ROS 2 Sensor-Fusion Architecture:** A hardware-independent, ROS 2-native SLAM architecture integrating RGB-D and 2D LiDAR data through RTAB-Map for real-time indoor 3D reconstruction.
- **Digital-Twin Validation Methodology:** A digital-twin-based workflow enabling systematic validation of the

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complete SLAM pipeline in simulation prior to physical deployment.

- **Quantitative Reconstruction Evaluation:** A geometric validation framework based on Statistical Outlier Removal (SOR) filtering and C2M signed distance analysis to assess reconstruction precision.
- **Cross-Platform Deployment Demonstration:** Experimental demonstration that the identical ROS 2 software architecture executes without code modifications on both simulation and resource-constrained embedded hardware.

II. RELATED WORK

Recent SLAM research has investigated different sensing modalities to improve robustness in complex indoor environments. Visual SLAM approaches based on monocular, stereo, and RGB-D cameras achieve accurate local estimation through feature tracking or direct photometric information, but remain vulnerable to textureless regions, illumination variations, and dynamic occlusions [4], [5]. Conversely, LiDAR-based SLAM provides reliable geometric measurements and improved robustness against illumination changes, since LiDAR is an active sensor that emits laser pulses and measures their return, whereas cameras are passive and depend on ambient scene illumination. Nevertheless, 2D LiDAR is inherently limited to planar observations, while 3D LiDAR introduces higher computational requirements, increased cost, and payload constraints for compact robotic platforms [3], [5].

To overcome individual modality limitations, recent studies have explored heterogeneous sensor fusion. Hybrid RGB-D and LiDAR approaches improve localization robustness by combining accurate planar constraints from laser measurements with dense depth information from visual sensors. For instance, 2.5D SLAM strategies integrate 2D LiDAR with RGB-D observations to improve obstacle representation and reduce failures from incomplete geometric perception [3]. Similarly, tightly coupled frameworks combining wheel odometry, inertial measurements, visual data, and laser scans achieve improved long-term localization [8]–[10]. Despite these advances, many approaches rely on hardware-specific configurations or tightly coupled optimization schemes, limiting portability across physical and virtual platforms.

Beyond sensor selection, the quality of the estimated motion model directly influences SLAM backend performance. Wheeled mobile robots are affected by non-holonomic constraints, wheel slippage, and terrain irregularities, causing accumulated odometric errors. While filtering-based approaches such as Extended Kalman Filters (EKF) have been widely studied [11], this work relies on scan-matching refinement based on the Iterative Closest Point (ICP) algorithm [7]. When constrained to the planar degrees of freedom of ground vehicles, ICP refinement provides improved numerical stability while reducing computational requirements.

Among open-source SLAM frameworks, RTAB-Map has been extensively adopted due to its capability of integrating heterogeneous sensor streams, performing graph-based optimization, and managing large-scale environments through memory-efficient loop closure strategies [7], [12]. Recent extensions have investigated semantic information and deep-learning-based front-end modules to improve robustness against dynamic objects and unstable visual features [13]. However, most RTAB-Map-based implementations are evaluated within specific robotic platforms or datasets, with limited investigation of software portability and direct sim-to-real transitions.

In addition to localization performance, quantitative validation of reconstructed environments has gained importance, particularly for digital twins and geometric inspection. Point clouds from depth and LiDAR sensors often contain measurement noise and isolated outliers, requiring post-processing such as Statistical Outlier Removal (SOR) filtering [14]. Following noise reduction, Cloud-to-Mesh (C2M) analysis evaluates reconstructed geometry by computing the distance from each point in the reconstructed cloud to the nearest triangle of a reference mesh. In CloudCompare, these distances can be signed according to the orientation of the reference triangle normals [15]. Nevertheless, many SLAM studies still rely primarily on trajectory-based metrics or qualitative map inspection, while fewer perform systematic geometric validation against high-fidelity digital-twin references.

Therefore, although recent research has significantly advanced multimodal SLAM, a gap remains regarding portable, ROS 2-native architectures that combine heterogeneous sensing, direct simulation-to-real deployment, and rigorous quantitative geometric validation within a unified framework. This work addresses this gap by presenting a portable RGB-D and 2D LiDAR fusion architecture that maintains software consistency across environments and evaluates reconstructed geometry using a digital-twin-based C2M analysis.

III. METHODOLOGY AND EXPERIMENTS

This section describes the experimental infrastructure and the theoretical foundation of the proposed SLAM system.

A. Simulation Environment and Computational Infrastructure

The simulation environment was modeled in Blender and executed in Gazebo Classic 11, natively integrated into ROS 2 (Humble) under Ubuntu 22.04 LTS. The virtual scenario reproduces a military transport aircraft based on an existing physical structure at the Department of Aeronautical Engineering, São Carlos School of Engineering (EESC-USP). This geometric correspondence (Fig. 1) was designed to enable future comparisons between real and simulated conditions.



Fig. 1. Experimental environment. (a) Real environment. (b) Digital twin used in the experiments.

The experiments were processed on an Intel Core i5-9400F CPU and an NVIDIA GeForce GTX 1060 GPU, a configuration sufficient to sustain the sensors nominal sampling rates without degrading the processing pipeline.

B. BART Robotic Platform

The robotic platform was the BART (Basic Autonomous Rover for Testing), developed by the AeroTech Laboratory (EESC-USP). As shown in Fig. 2, the platform features a two-tiered rectangular base of 15.9 cm x 12.5 cm, with omnidirectional kinematics driven by four 6.5 cm diameter Mecanum wheels.

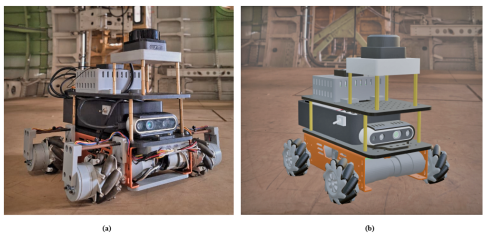


Fig. 2. BART platform. (a) Real Representation. (b) Digital twin used in experiments.

Its perception system comprises two sensors connected to a Toradex Verdin iMX8M Plus System on Module (SoM), with readings integrated via ROS 2:

(i) an Intel RealSense D435i RGB-D camera, operating at 15 Hz with a nominal horizontal field of view of 1.2354 rad ($\approx 70.8°$);
(ii) an RPLiDAR C1 sensor, configured for 360° scans at 10 Hz, with a maximum range of 10 m and an angular resolution of 460 points per revolution.

Although the sensors operate at their nominal rates, RTAB-Map registers keyframes at a configurable detection rate rather than at the full sensor rate, which avoids redundant graph updates (Section III-D).

C. Operational Architecture

The integration and processing of sensor data are shown in Fig. 3. The data stream operates synchronously, ensuring consistency between sensor readings and odometry. Topics from the RGB-D camera, LiDAR, and BART's odometry are forwarded to the central node based on RTAB-Map [7]. This same node operates in both the simulation environment and on the Toradex Verdin embedded platform, preserving its functional architecture and requiring only adjustments to external parameterization.

This node handles the computationally intensive mapping process, grounded in a graph-based SLAM approach [16], [17]. It employs incremental memory management that organizes graph nodes into short-term and long-term memories, reducing the search space during loop closure detection and enabling real-time responses [13].

In the pipeline (Fig. 3), motion tracking relies on kinematic odometry combined with visual feature estimation from the

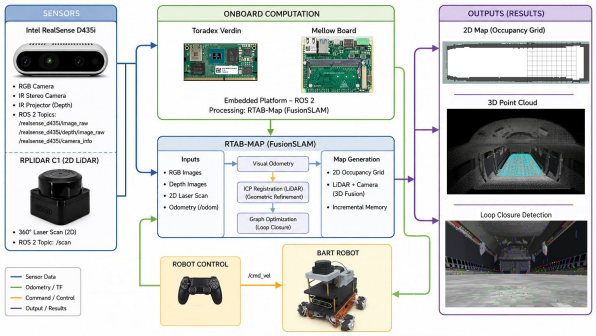


Fig. 3. Proposed Sensor Fusion SLAM Pipeline.

RGB-D sensor to update the pose graph. In parallel, 2D LiDAR data generates the 2D occupancy grid via ray tracing for obstacle avoidance and planar navigation, while providing scan registration constraints when matching consecutive sweeps. Once the pose graph is optimized, dense depth images from the RGB-D camera are back-projected to construct the final 3D point cloud. The pipeline produces three main outputs:

(i) a 2D occupancy grid map for navigation;
(ii) a global, optimized 3D point cloud used for geometric evaluation;
(iii) the loop closure logs.

D. Experimental Protocol

The algorithm processes graph updates at a detection rate of 1 Hz, decoupled from the sensor acquisition rate. This configuration reduces the computational burden, limits redundant graph updates, and decreases the likelihood of false-positive loop closures. New keyframes are further constrained by translational and rotational thresholds, so not every update creates a new pose-graph node.

As a result, although the RPLiDAR C1 operates at 10 Hz (40,570 raw scans over the 4,057 s experiment), only 3,952 scans were processed as keyframes. This intentional reduction reflects the decoupling between sensor acquisition and graph updates.

To evaluate the proposed SLAM system, BART was commanded to perform a complete reconnaissance run inside the simulated aircraft cabin, following a predefined trajectory designed to ensure sufficient overlap for loop closure detection. The operational parameters and quantitative characteristics of the run are detailed in Table I.

TABLE I
EXPERIMENTAL PROTOCOL PARAMETERS

Parameter	Value	Unit
Robot linear speed	0.12	m/s
Experiment duration	4,057	s
Travel distance	380.67	m
LiDAR scans processed as keyframes	3,952	–
Keyframes inserted in Working Memory	865	–
Detected loop closures	338	–

During the experiment, the robot's linear speed was kept at approximately 0.12 m/s during traversal segments to prevent excessive motion blur in the RGB-D camera and to ensure consistent LiDAR scan registration. The complete path length of 380.67 m was traversed in 4,057 s. Throughout this period, the RPLiDAR C1 generated a total of 40,570 raw scans, of which 3,952 were processed as keyframes by the RTAB-Map pipeline, resulting in 865 keyframes inserted into the graph. A total of 338 loop closures were successfully detected and optimized, correcting the accumulated drift along the trajectory.

E. Mathematical Foundations

The mathematical foundation rests on two pillars: the fusion algorithm and the statistical methods used for spatial validation of the map.

1) Fusion and Optimization Algorithm: The initial pose estimate comes from the odometry covariance matrix Q given in (1), parameterized by the locomotion system's uncertainty model. This matrix [11] models the uncertainties in the translational and rotational components used for initial pose prediction.

$$Q = \text{diag}(\sigma_x^2, \sigma_y^2, \sigma_\theta^2) \tag{1}$$

where σ_x^2 and σ_y^2 are the variances of the linear translation noise, and σ_θ^2 is the angular noise variance.

This prediction is refined by RTAB-Map's ICP algorithm, which performs rigid registration between consecutive LiDAR scans, estimating the planar transformation that best aligns the point clouds, as in (2).

$$E(x, y, \theta) = \sum_{k=1}^{N} \left\| P_{k,i-1} - \left(R(\theta) P_{k,i} + \begin{bmatrix} x \\ y \end{bmatrix} \right) \right\|^2 \tag{2}$$

where $P_{k,i}$ and $P_{k,i-1}$ are corresponding points from the current and reference LiDAR scans, $R(\theta)$ is the rotation matrix for the robot's orientation, and $\begin{bmatrix} x & y \end{bmatrix}^T$ is the planar translation vector [18].

The ICP-corrected poses are then added to RTAB-Map's pose graph. When a loop closure is detected, the trajectory is optimized by minimizing the global cost function $F(\mathbf{x})$ in (3), using the information matrix Ω_{ij} (inverse of the alignment error covariance matrix) [19].

$$F(\mathbf{x}) = \sum_{(i,j)} e_{ij}^T \Omega_{ij} e_{ij} \tag{3}$$

2) Filtering and Computational Metrics: After global pose graph optimization and 3D map consolidation, post-processing was performed in CloudCompare. To remove noise and isolated points, the Statistical Outlier Removal (SOR) filter was applied based on local density consistency. For each point (P_i), the mean distance ($\bar{d}_i$) to its k nearest neighbors (P_j) is computed as in (4) [14]:

$$\bar{d}_i = \frac{1}{k} \sum_{j=1}^{k} \text{dist}(P_i, P_j), \quad \text{where} \quad \bar{d}_i > \mu + m \cdot \sigma \tag{4}$$

where μ and σ are the mean and standard deviation of the global distance distribution, and m is the filter's sensitivity factor configured in the software.

The map is then evaluated quantitatively and qualitatively via the Cloud-to-Mesh (C2M) metric, which computes the minimum signed Euclidean distance between the post-processed point cloud and the original aircraft mesh (ground truth) built in Blender. Residual errors are assessed using MAE and RMSE, followed by a goodness-of-fit analysis to a theoretical normal distribution via a frequency histogram.

IV. RESULTS AND DISCUSSION

The experimental validation was structured into two consecutive stages. First, an analysis of the system performance of the mapping algorithm is presented, emphasizing both the temporal and statistical stability of the processing rates. Next, the residual metric errors relative to the reference three-dimensional model are quantified.

A. System Performance and Geometric Validation

The system's performance was evaluated based on temporal metrics extracted directly from the `rtabmap.db` database, as well as spatial and accuracy metrics computed through post-processing in CloudCompare during the experimental trials. The obtained results are presented in Table II.

TABLE II
SYSTEM PERFORMANCE METRICS

Parameter	Value	Units
Memory update	148.49	ms
Map optimization	0.006	ms
Timing total	153.24	ms
Standard Deviation	0.0439	m
MAE	0.0308	m
RSME	0.0536	m

As evidenced in Table II, and given the 1 Hz detection rate established in Section III-D, each processing cycle has a time window of up to 1000 ms. The total processing time of 153.24 ms demonstrates that the system processed the synchronous sensor stream with an 84.7% temporal margin relative to the critical real-time limit. Even though most of this execution time was concentrated on node updates (corresponding to 97%), the residual time required for global graph optimization remained nearly instantaneous, confirming that the computational infrastructure sustained the nominal sensor rates in real time without pipeline degradation. These metrics, however, derive from a single simulated run and are indicative rather than generalized.

Simultaneously, the spatial consistency of the reconstructed point cloud was evaluated through a visual comparison with the reference 3D model (ground truth) of the military aircraft, as outlined in the workflow of Fig. 4.

The raw mapping output (Fig. 4b) initially yielded $6,682,407$ points. To eliminate isolated outliers and spurious reflections typical of the virtual environment, a Statistical

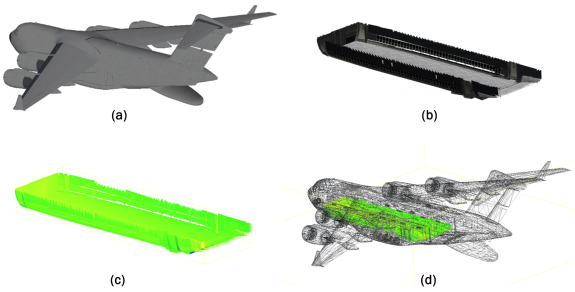


Fig. 4. Workflow adopted for geometric validation of the reconstructed point cloud. (a) original polygonal mesh. (b) raw point cloud. (c) point cloud after noise removal. (d) overlay of final reconstruction and reference model.

Outlier Removal (SOR) filter was applied. This process removed $385,784$ points, corresponding to 5.77% of the original dataset, resulting in a filtered point cloud of $6,296,623$ points (Fig. 4c). This step successfully preserved the geometry of the aircraft's internal walls and floor, as shown in the overlay comparison in Fig. 4d.

From this consolidated point cloud, the C2M distance was computed using the ground truth mesh as the baseline for comparison. The resulting metrics, compiled in Table II, indicate a high geometric agreement with the 26.7 m-long military cabin, achieving an accumulated error of approximately 0.115% based on the Mean Absolute Error (MAE) of 0.0308 m and a Root Mean Square Error (RMSE) of 0.0536 m.

In the physical deployment (Fig. 5), the architecture executed reliably on the Toradex Verdin SoM, maintaining real-time processing and stable time synchronization across heterogeneous streams. The 2D LiDAR generated the occupancy grid and planar boundaries for navigation, while the RGB-D front-end anchored the 3D reconstruction. Although LiDAR does not directly correct Mecanum wheel slippage, its geometric constraints prevent map inconsistency in repetitive fuselage sections where visual features alone become ambiguous. However, no ablation study isolating the contribution of ICP refinement or of each sensor modality was conducted; therefore, the marginal benefit of LiDAR-based refinement over the RGB-D pipeline alone cannot be quantified from the present results.

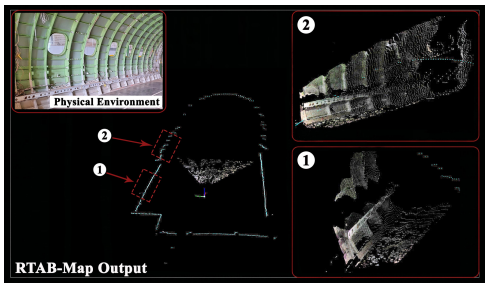


Fig. 5. Pipeline deployment in a real-world environment.

The physical experiment provides a qualitative demonstration of the mapping pipeline operating on the BART platform. No quantitative geometric evaluation against a real-world ground-truth reference was performed. Therefore, deployment in both environments demonstrates operational feasibility, but does not establish equivalent reconstruction accuracy between simulation and the physical robot. Quantitative validation of real-world mapping accuracy remains a limitation of this study and a direction for future work.

B. Statistical Evaluation and Error Validation

To statistically characterize the metric deviations of the reconstruction, CloudCompare was utilized to generate the frequency histogram of the signed C2M distances. The experimental data were compared against a theoretical Gaussian distribution (with $\mu = 0.0307$ m and $\sigma = 0.0439$ m), as presented in Fig. 6.

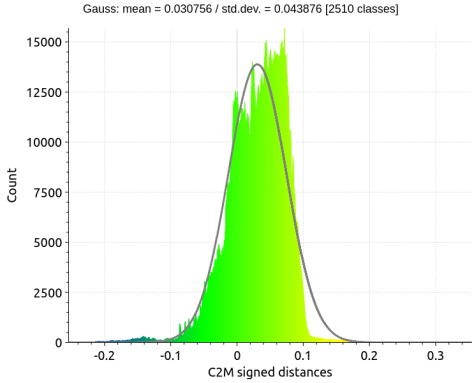


Fig. 6. CloudCompare statistical analysis.

The residual distribution exhibits an approximately Gaussian behavior with slight negative skewness, characterized by a tail extending to approximately -0.20 m. Despite this localized deviation, the residuals remain concentrated around low-magnitude values, indicating the absence of dominant systematic errors. Combined with the geometric validation, these results indicate that, within this single simulated run, the integration of pose graph optimization, ICP refinement, and loop closure effectively mitigated accumulated drift during mapping.

The obtained geometric accuracy is comparable to recent visual and Lidar SLAM approaches, such as PLPM SLAM [20] and LRSLAM [21], which employ additional constraints to improve reconstruction quality. In contrast, the proposed RGB-D and LiDAR fusion pipeline achieves centimeter-level consistency in large-scale indoor environments using a conventional graph-based optimization framework, demonstrating a favorable trade-off between accuracy and system complexity.

V. CONCLUSION

This work presented a portable ROS 2-based sensor fusion architecture for indoor 3D reconstruction by integrating RGB-D and 2D LiDAR data through RTAB-Map. The same software architecture was maintained across simulation and

physical deployment, enabling validation through a digital twin methodology.

In a single simulated run, the system achieved real-time operation (153.24 ms per cycle, 84.7% margin relative to the 1 Hz requirement) and centimeter-level geometric accuracy against the reference model (MAE = 0.0308 m, RMSE = 0.0536 m), confirming the effectiveness of ICP refinement and pose graph optimization in reducing mapping error within the simulated environment.

The physical deployment on BART confirmed functional and operational portability under embedded resource constraints. However, no quantitative ground-truth comparison was performed on the physical platform, and the reported metrics derive from a single simulated run without repeated trials. Additionally, no ablation study isolating the contribution of ICP or of each sensor modality was conducted; therefore, the marginal benefit of LiDAR-based refinement cannot be quantified from the present results.

Future work will integrate an onboard inertial sensor into an EKF to stabilize odometry prior to graph insertion, conduct systematic ablation experiments on the same simulated trajectory, and perform repeated quantitative ground-truth validation inside the physical aircraft fuselage under diverse lighting conditions.

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